TURTLE

Unified Digital Learning Platform

n8n | Bilimclass | HTML/JS/CSS | Webhooks

The Problem

No Unified Digital Environment

Scattered Resources

Study materials spread across multiple platforms with no central hub

No AI Assistant

Students lack instant, intelligent support during self-study

Missed Schedules

No unified calendar or schedule tracking for lyceum events

Solution: lyceum portal that unifies schedules, resources, news & an AI chatbot in one place.

PROTOTYPE

Key Features

News Feed

Latest lyceum announcements and events in real time

Schedule

Weekly timetable with class and event integration

AI Chatbot

Instant answers to student questions via Groq AI

Resources

Centralized library of study materials and links

Student Portal

Personal dashboard for each student

Complaint System

Planned: anonymous feedback & issue reporting

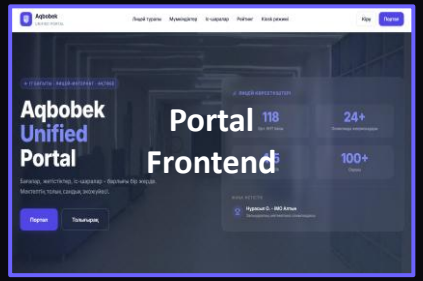
AI CHATBOT

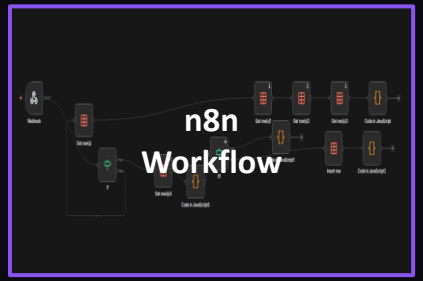
Smart Student Assistant

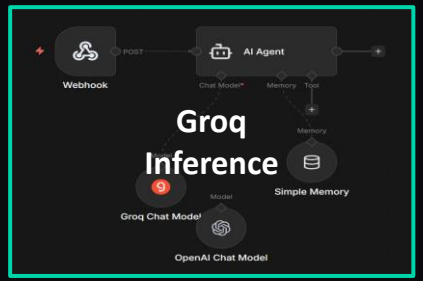
Powered by Groq via inference layer. Students ask questions - the AI answers instantly in context.

1. Student sends a message via the portal chat
2. Webhook triggers n8n automation workflow
3. n8n calls Groq API with Claude model
4. Response returned and displayed in real time

How It Works







Response flows back through n8n → Portal

TECHNOLOGY

Tech Stack

Frontend

HTML
Markup & structure

CSS
Styling & responsive layout

JavaScript
Interactivity

JSON
Data exchange format

Backend & AI

n8n
Workflow automation engine

Webhooks
Event-driven triggers

Groq
Fast AI inference layer

Javascript
Logic

ROADMAP

What's Next?

If we had one more month...

Communication System

Phase 2

Built-in messaging between students and teachers

Complaint & Feedback

Phase 2

Anonymous issue reporting to school administration

Mobile App

Phase 3

Native iOS & Android apps for on-the-go access

Personalized AI Tutor

Phase 3

Adaptive learning paths based on student performance

TEAM

Meet the Builders

Нұрасыл Оралбеков

Backend Developer

Designed and built the n8n automation workflows, webhook integrations, and AI pipeline connecting Groq.

Аружан Маратова

Frontend Developer

Built the portal UI - HTML, CSS, JavaScript making the platform intuitive and responsive for students.

Адема Тилишбаева

Designer

Created the visual identity, branding, slide design, and overall UX of the TURTLE platform.